

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

*I Ebin Antony P (192373011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

CONCEPT MAPPING

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

- a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

- b. Show how each concept is mathematically connected
- c. Map their role in encryption/decryption
- d. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

3. Classical Encryption Schemes

Substitution Techniques $\longleftrightarrow$ Transposition Techniques

Substitution Techniques

Replace each plaintext symbol (letter, digit, or character) with another symbol based on a fixed rule or key.

Eg: Caesar Cipher, Monoalphabetic Substitution, Playfair cipher.

Similarities

- Both are classical (pre-computer) encryption methods.
- Both operate on plaintext using a key (even if simple) $\longleftrightarrow$
- Both provide confidentiality by transforming the original message.
- Both can be broken using frequency analysis and other cryptanalytic techniques (if weak).

Transposition Techniques

Rearrange (permute) the positions of the plaintext symbols without changing symbols themselves.

Eg: Rail Fence Cipher, Columnar Transposition, Route Cipher.

Effect on Data Structure

- Alters actual symbols in data.
- Length & overall structure usually remains same.
- Example:

Plaintext: HELLO

Ciphertext: KHOOR (Caesar shift #3)

→ Same length, different symbols

Effect on Data Structure

- Does not change symbols themselves.
- Alters orders/position of symbols.
- Length & character set remain same.
- Eg: Plain text: HELLO
Ciphertext: HELLO (Simple transposition)
→ Same symbols, different order.

Effect on Security Level

- Provides moderate security
- Hides meaning of message by changing symbols
- Vulnerable to frequency analysis (letter occurrence patterns).
- Stronger only if a polyalphabetic or well-designed substitution is used

Effect on Security Level

- Provides lower security than substitution (in general)
- Does not hide symbol frequency (letters are still the same)
- Easier to analyze using known plaintext or pattern recognition.
- Becomes stronger with complex permutations (e.g. columnar with key)

Which is More Vulnerable to Attacks? → Transposition Techniques

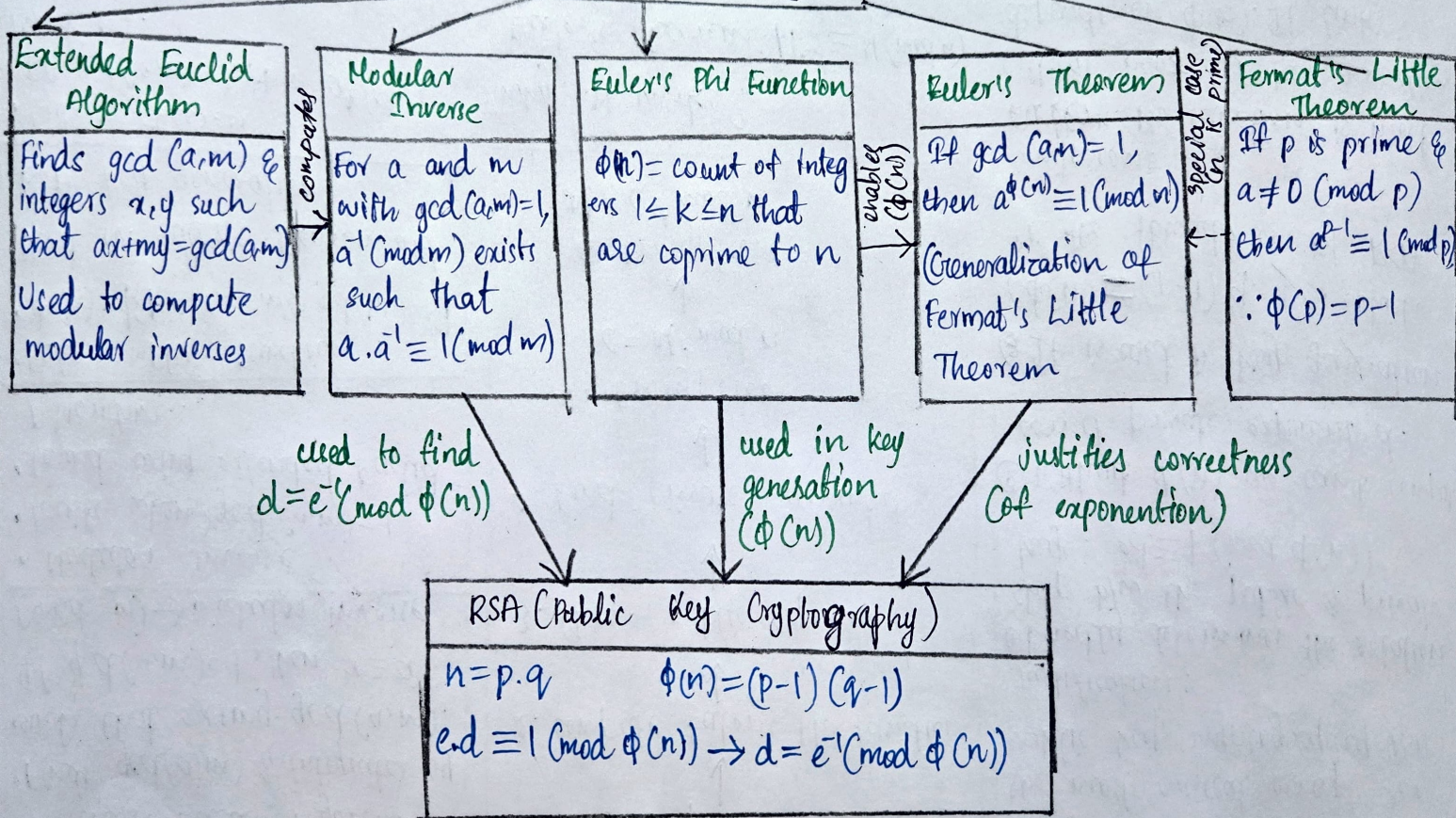
Why Transposition is More Vulnerable

- It does not change the letter frequencies, so frequency analysis can still reveal common letters (e.g., E, T, A, O).
- With enough ciphertext and a bit of pattern analysis, the original order can often be reconstructed
- Known-plaintext attacks are easier because the symbols are changed (only ^{reordered} symbols)
- Simple transposition (e.g., rail fence) can be often broken with minimal effort.

Why substitution is More Resistant?

- It hides the actual symbols, making frequency analysis less direct.
- Without knowing the key, it's harder to map ciphertext to plaintext.
- However, monoalphabetic substitution is still vulnerable to frequency analysis and cryptanalysis (e.g., guessing the key or using known patterns).
- Polyalphabetic substitution (e.g., Vigenere) significantly increases resistance compared to simple substitution.

5. Number Theory Concepts in Cryptography



a. Mathematical Connections

- Extended Euclid Algorithm
- Finds $\gcd(a, m)$ & integers x, y such that $ax + my = \gcd(a, m)$. If $\gcd(a, m) = 1$, then $x \equiv a^{-1} \pmod{m} \rightarrow$ modular inverse

- Modular Inverse
- Exists when $\gcd(a, m) = 1$.
- Found using extended Euclid Algorithm.

- Euler's Phi Function
- $\phi(n) = |\{k \in \{1, \dots, n\} : \gcd(k, n) = 1\}|$
- Used in Euler's Theorem and RSA key generation.

- Euler's Theorem
- If $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.
- For prime n , reduces to Fermat's little theorem $C \phi(p) = p - 1$

b. Role in Encryption/Decryption

1. Choose primes p, q
 $\downarrow$
2. Compute Euler's Phi Function
 $\downarrow$
3. Choose public exponent e
 $\downarrow$
4. Find private exponent d
 $\downarrow$
5. Encryption
$$C = M^e \pmod{n}$$

 $\downarrow$
6. Decryption
$$M = C^d \pmod{n}$$

Why it works?

Euler's Theorem: $M^{\phi(n)} \equiv M \pmod{n}$

c. Most Critical Concept

Euler's Phi Function ($\phi(n)$) is the most critical concept for Public key cryptography (RSA).

Justification:

- ① Directly determines the relationship b/w the public & private keys: $ed \equiv 1 \pmod{\phi(n)}$
- ② Without $\phi(n)$, we can't compute correct private exponent d .
- ③ It is used in key generation $\phi(n) = (p-1)(q-1)$, which is the foundation of RSA.

Conclusion:

Euler's Phi Function is most critical bcoz RSA key generation depends on $\phi(n)$. It links public and private keys.